Laboratory 1
Hardness Testing

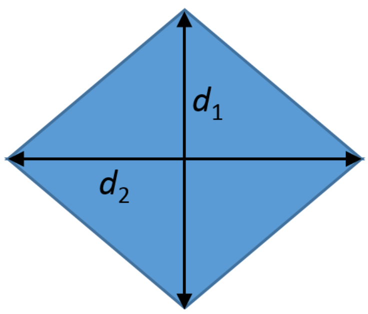
1) Introduction

An important material property that is frequently needed is *hardness*, which is essentially a measure of a material's resistance to indentation when a highly localized force is applied to its surface. For materials such as metals, localized indentation leads to plastic deformations that are permanent, whereas for polymeric ones such as rubbers and elastomers the deformation may be inelastic (e.g. viscoelastic) but it is fully recoverable. In this experiment, the hardness of specimens from both classes of materials will be measured, each having significantly different underlying physical mechanisms of deformation and consequently different theoretical background for interpretation as well as varying implications regarding material selection in design. For further information, you will be referred to sections of ASM Handbook Vol.8 on "Mechanical Testing and Evaluation" (see https://dl.asminternational.org/handbooks/book/47 which is available online through the campus network).

2) Test Method

2.1) Metals

For metals, using the micro-hardness machine, the so-called *Vickers* test is going to be performed on a steel specimen to measure its *Vickers Hardness* (HV or VHN) value. Specifically, a small indentor in the shape of a right pyramid and made out of diamond is pressed onto the surface of the material to be measured. The hardness measurement involves the observation of the resulting impression of the indentor on the material surface under an optical microscope, the dimensions of which are then utilized to determine the *Vickers Hardness* value for the material using the formula $HV = \frac{1.8544\,P}{d^2}$ where P is the applied normal load (kgf) and $d = (d_1 + d_2)/2$ (see drawing below – dimensions in mm). Higher HV values indicate higher resistance to plastic deformation. HV should therefore conceptually correlate with the ultimate tensile strength of the material. For further details regarding the testing procedure, you may consult ASM Handbook Vol.8 > Hardness Testing > Microindentation Hardess Testing > Vickers Hardness Test. An additional source is Dowling, section 4.3.2 (4th Ed: 4.7.2). The test itself is governed by the ASTM E92 test standard.



2.2) Polymers

For polymers, the so-called *Shore* hardness test is going to be performed on selected porous and solid specimens with low elastic moduli using a *durometer*. For such materials, the *A-scale* of Shore hardness that is governed by the ASTM D2240 test standard is appropriate. Because the deformations are not permanent in this test, the desired piece of information that one primarily wishes to extract is the Young's modulus of the material. For further details regarding this test, you may consult ASM Handbook Vol.8 > Hardness Testing > Miscellaneous Hardness Tests > Durometer Hardness Testing. Within this test, an indentor penetrates the specimen by an amount w (mm) with a load F. The relation between w and F may be derived theoretically using elasticity theory under the following assumptions: *the material is isotropic and elastic, the specimen is infinitely large with respect to the indentor dimensions, the indentor is cylindrical with diameter* d *and its contact with the specimen is frictionless*. When these hold, the relation is $w=\frac{F(1-\nu^2)}{dE}$ where ν is the Poisson's ratio (almost 0.5 for most polymers, which are nearly incompressible) and E (MPa) is the unknown Young's modulus[1]. However, the durometer does not indicate w or F but rather displays the *Shore hardness value* s. Hence, additional information[2] is needed to extract E:

1. The reading s from the durometer is such that a full penetration (2.5mm) corresponds to a value of $s=0$ (indicating no resistance to deformation) and no penetration corresponds to $s=100$ (indicating extreme resistance to deformation): $w=C_3(100-s)$
2. The indentor is connected to a *preloaded* spring within the durometer such that there is a simple linear relation between w and the change in F, governed by a spring constant. In view of item 1, a similar relation therefore holds between s and F: $F=C_1+C_2 s$.

Above, $(d, C_1, C_2, C_3)=(0.79\,mm, 0.549\,N, 0.07516\,N, 0.025\,mm)$. Combining all of this information together, one may finally obtain a relation between the durometer reading and the Young's modulus as desired[3]: $E(s)=\frac{3}{4dC_3}\frac{C_1+C_2 s}{100-s}$. Notice that, among other issues (such as whether all the assumptions are satisfied in reality), there are two problems with this relation:

1. $s=0$ will be displayed for all materials with Young's modulus less than $E(0)$ so there is a minimum hardness that delivers a meaningful result (or, the *Shore 00* scale may be used).
2. $s\to 100$ does indicate $E\to\infty$ but *Shore A* is not designed for very stiff materials to begin with. Instead, one should move to another test scale (*Shore D*, for instance).

Hence, in practice, measurements near the extreme ends of the range $s\in[0,100]$ should be met with suspicion. Otherwise, however, the stated expression is one relation (among others[3]) which delivers values that faithfully indicate independent measurements of the Young's modulus[2].

[1] S.P. Timoshenko, J.N. Goodier. *Theory of Elasticity.* 3rd Edition. 1970.

[2] J. Kunz, M. Studer, *Determining the Modulus of Elasticity in Compression via the Shore A Hardness*, Kunststoffe International, 6:1-3, 2006.

[3] A.W. Mix, A.J. Giacomin. *Dimensionless Durometry.* Polymer-Plastics Technology and Engineering, 50:288-296, 2011.

3) Experiment

3.1) Metals

The hardness test will be performed on three specimens: *steel*, *copper* and *aluminum*. However, the exact material specifications for each specimen are not known.

(a) Make one measurement of Vickers hardness on each specimen using *P=1 kgf*. Record each value. Using the collected data from all groups (*to be sent out*) calculate the mean and the standard deviation for each specimen. Comment on possible reasons for the variation in measurement results.

(b) Based on d_1 and d_2 for one of the measurements, check if the measurement agrees with the equation for *HV*.

(c) Based on a literature search, determine which type of steel specimen was possibly employed.

(d) Another common indentation-based hardness test is the so-called *Brinell* test. Using an appropriate conversion table, calculate the mean *Brinell Hardness* (HB) for the test specimen. An empirical relation allows one to relate the measured *HB* value to the ultimate strength of a steel specimen: $\sigma_u = 3.45 \times HB$. Calculate the ultimate strength σ_u of the test specimen. Is this consistent with your guess in part (d)?

(e) When would you desire a metal with a high or low hardness? Also, make an online search on how to manufacture a mechanical part which has high hardness on the outside but low hardness on the inside and why it would be desirable to manufacture such a part.

3.2) Polymers

The hardness test will be performed on three different *polymeric* specimens. However, the exact material specifications for each specimen are not known.

(a) Using the Shore durometer, make one measurement of Shore A hardness for each specimen. Record each value. Using the collected data from all groups (*to be sent out*) calculate the mean and the standard deviation for each specimen. Comment on possible reasons for the variation in the measurement results as well as on the reliability of the results.

(b) Convert the Shore A hardness readings to Young's moduli values using the relation provided earlier. Also, plot this relation and answer the following question: do small errors in hardness readings correspond to larger or smaller errors in Young's moduli values with increasing hardness?

(c) Make an online search for typical Shore A hardness values and which materials they possibly correspond to. Do these general observations correlate with your findings from part (b)?

(d) Carefully inspect the materials and the testing apparatus. Then, list three reasons why the Young's modulus determined via the relation provided earlier may not match the one from a tension test.

(e) Evaluate $E(0)$ and subsequently carry out an online search to find at least two materials which cannot be characterized faithfully through the Shore A hardness test.